SUNY Old Westbury 2026-2027 Undergraduate Catalog

Computer and Information Science, B.S.

HEGIS Number: 0701

Curriculum Code: U0286

Requirements for the Bachelor of Science (B.S.) Degree in Computer & Information Science

The Computer & Information Science (CIS) degree program focuses on the subjects of computers, their organization, and the programs that run them. Emphasis is placed on software rather than hardware. CIS majors may choose to specialize in either **system software** or **information systems** or **computer and network security** by appropriate selection of CIS electives. See sample programs at end of section.

CIS Major Requirements: 68 Credits (17 courses)

1. Required Courses: 40 Credits (10 courses)

- CS 2510 - Computer Programming I Credits: 4
- CS 2511 - Computer Programming II Credits: 4
- CS 3620 - Computer Architecture I Credits: 4
- CS 3810 - Data Structures and Algorithms Credits: 4
- CS 4100 - Technical Communications Credits: 4
- CS 4501 - Software Engineering Credits: 4
- CS 4550 - Database Management Systems Credits: 4
- CS 4720 - Internet and Web Technologies Credits: 4
- CS 5910 - System Design & Implementation Credits: 4

2. Required Elective courses: 12 Credits (3 courses)

Take any three of the following courses:

- CS 4400 - Artificial Intelligence Credits: 4
- CS 4705 - Introduction to Computer Security Credits: 4

- CS 4710 - Applied Cryptography Credits: 4
- CS 5610 - Operating Systems Credits: 4
- CS 5710 - Computer Networks Credits: 4
- CS 5730 - Computer Network Security Credits: 4
- CS 5810 - Data Mining I Credits: 4

3.Required Mathematics Courses: 16 Credits (4 courses)

- MA 2310 - Calculus & Analytic Geometry I Credits: 4
- MA 3030 - Discrete Mathematics Credits: 4
- MA 3210 - Introduction to Probability and Statistics Credits: 4

- MA 3160 - Linear Algebra Credits: 4

  OR

- MA 4100 - Number Theory Credits: 4

Liberal Education Curriculum Courses

Refer to the Liberal Education Curriculum Bulletin for requirements.

General Electives

In consultation with academic advisor, for a total of 120 credits.

Other Requirements

- A grade of C or higher is needed in all required computer science and mathematics courses.
- Transfer students must complete a minimum of 28 credits (7 courses) of the required major courses at or above the 3000 level at Old Westbury.

College Wide Requirements

- 120 credits overall (40 credits at Old Westbury, may transfer a maximum of 80 credits)
- 45 Upper Division credits (3000, 4000, or 5000 level courses)
- 60 Liberal Arts credits
- Cumulative Grade Point Average of **2.0**

Sample Course Sequence By Major

Term 1

- MA 2090 - Precalculus Credits: 4
- CS 2510 - Computer Programming I Credits: 4

Term 2

- CS 2511 - Computer Programming II Credits: 4
- MA 3030 - Discrete Mathematics Credits: 4
- MA 2310 - Calculus & Analytic Geometry I Credits: 4

Term 3

- CS 3810 - Data Structures and Algorithms Credits: 4
- CS 3911 - C++ and Object-Oriented Programming Credits: 4
- MA 2310 - Calculus & Analytic Geometry I Credits: 4

Term 4

- CS 4501 - Software Engineering Credits: 4
- CS 4550 - Database Management Systems Credits: 4
- CS 3620 - Computer Architecture I Credits: 4

Term 5

- CS 4100 - Technical Communications Credits: 4
- CS 4720 - Internet and Web Technologies Credits: 4
- MA 3160 - Linear Algebra Credits: 4

  OR

- MA 4100 - Number Theory Credits: 4

Term 6

- CS Elective
- CS Elective

Term 7

- CS Elective

Term 8

- CS 5910 - System Design & Implementation Credits: 4